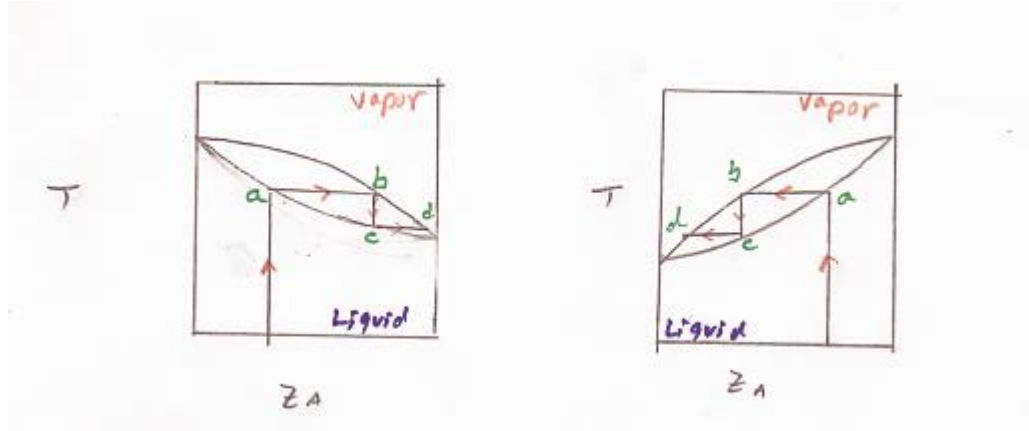


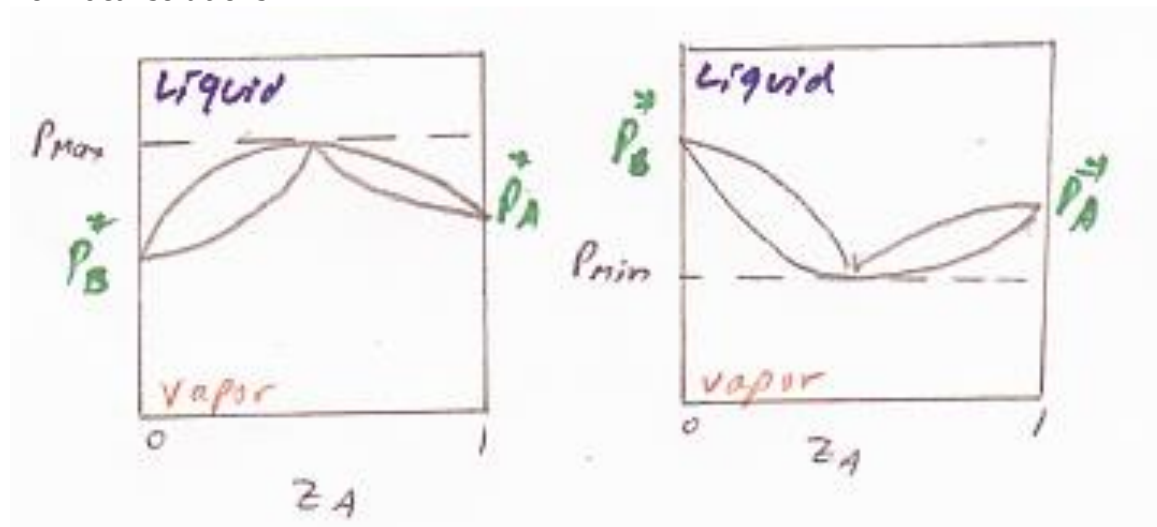
Distillation



x_A = mole fraction of A in the liquid
 y_A = mole fraction of A in the vapor
 z_A = overall mole fraction of A (for the entire system)

Point a. Mixture starts to boil, with $x_A = z_A$, $y_A = z_b$. Points b-c. Vapor is condensed to form a liquid with $x_A = z_b = z_c$. Point c. The liquid that was collected in the previous step is boiled to form a vapor with $x_A = z_d$. Condensation of the last bit of vapor produces a liquid very rich in either A (if $T_{b,A} < T_{b,B}$ on the left) or B (if $T_{b,A} > T_{b,B}$ on the right).

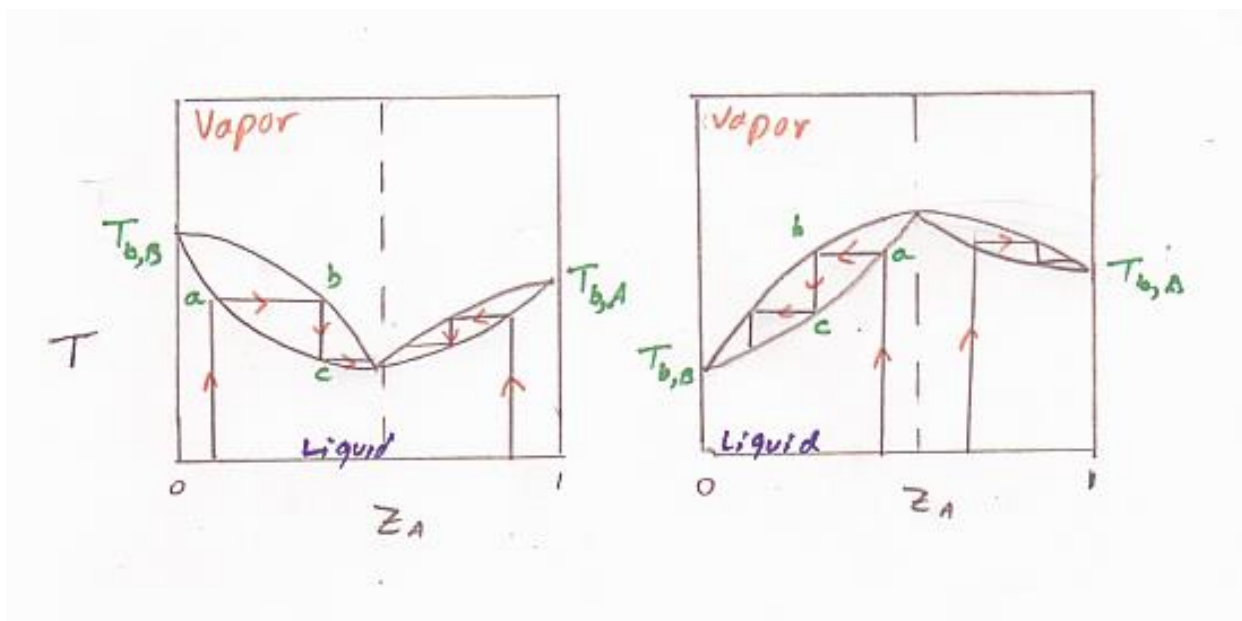
Non-ideal solutions



Left: Vapor pressure reaches a maximum because of repulsion between A and B. This is an azeotrope, where the liquid and vapor have the same composition.

(Right: Vapor pressure reaches a minimum because of attraction between A and B. This is also an **azeotrope**.

Distillation of non-ideal solutions



Left diagram: low boiling azeotrope. A solution of b composition z_a first boils at T_a . The vapor comes off with composition z_b . It is condensed and then boils at T_c . The final vapor to come off has the **azeotropic composition**, and the remaining liquid is enriched in B (or A, depending on which side of the azeotrope the process started).

Right diagram: high boiling azeotrope. As before, z_a first boils at T_a the final vapor to come off is enriched in B (or A), and the remaining liquid has the azeotropic composition.

TWO COMPONENT (BINARY) DIAGRAMS with $C = 2$

Composition is now variable: T, P, and composition can vary

$$P + F = C + 1$$

When vapor pressures are negligible and nearly constant:

Condensed phase rule:

$$P + F = C + 2$$

Pressure is no longer a variable: only T and composition matter

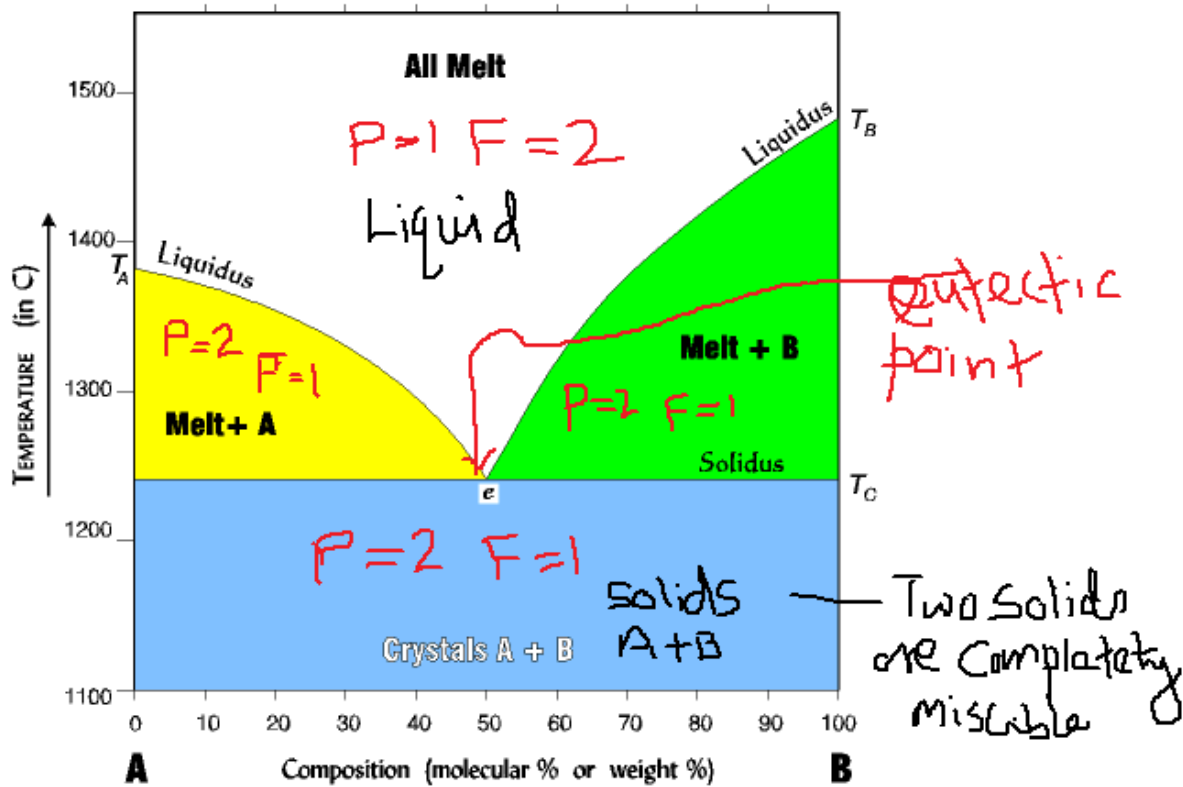
$$P + F = 3$$

Three possibilities (as before):

- $P = 1 \dots F = 2$ (bivariant phase field)
- $P = 2 \dots F = 1$ (univariant phase curve)
- $P = 3 \dots F = 0$ (invariant phase point)

SIMPLE EUTECTIC SYSTEMS

simplest binary diagram for solids



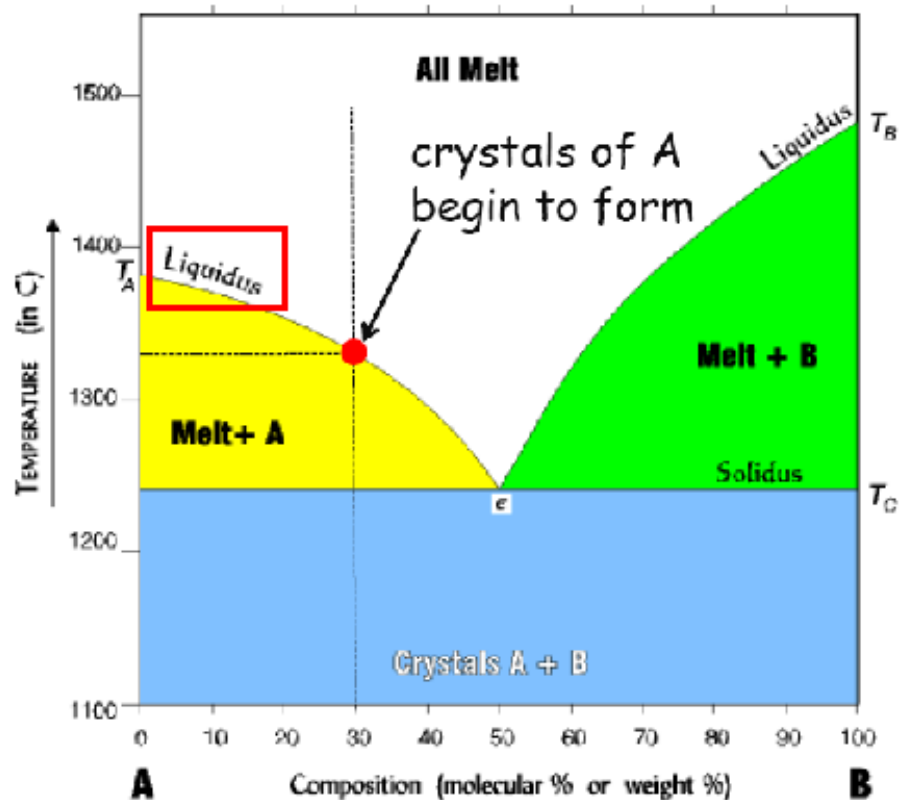
- no compounds/solid solutions in solid state, •only single phase liquid at high temperatures, •partial melting at intermediate temperatures

First, specify overall composition of the system

Second, pick a temperature.

→ The compositions of the phases (1 or 2) are then fixed

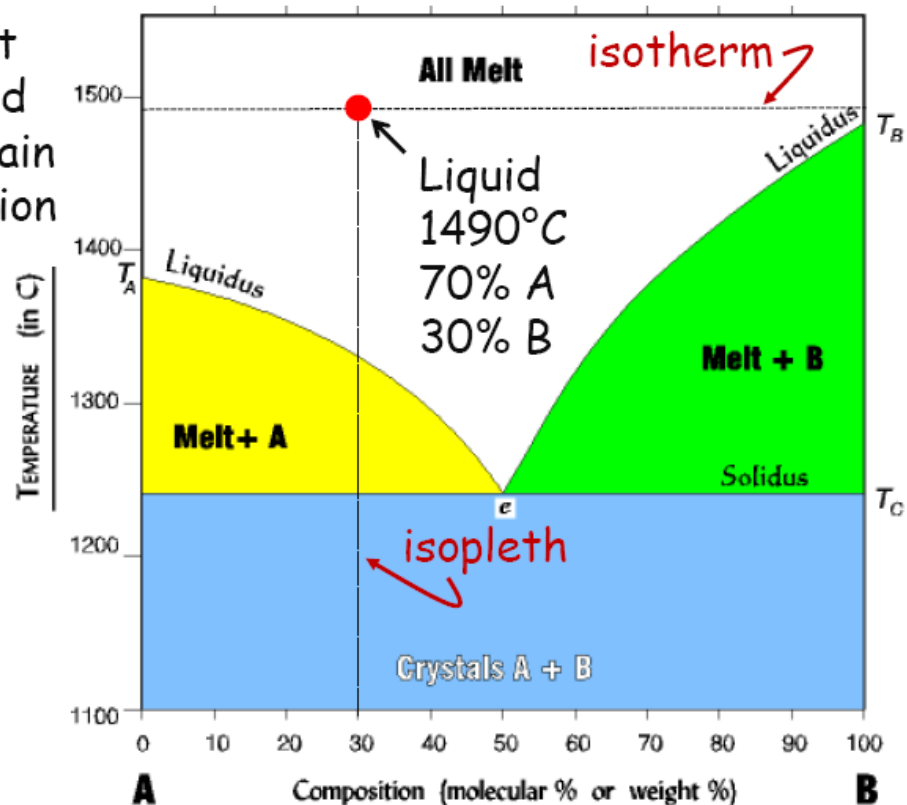
now, slowly
cool liquid



Liquidus curve: specifies the maximum temperature at which crystals can co-exist with the melt in thermodynamic equilibrium

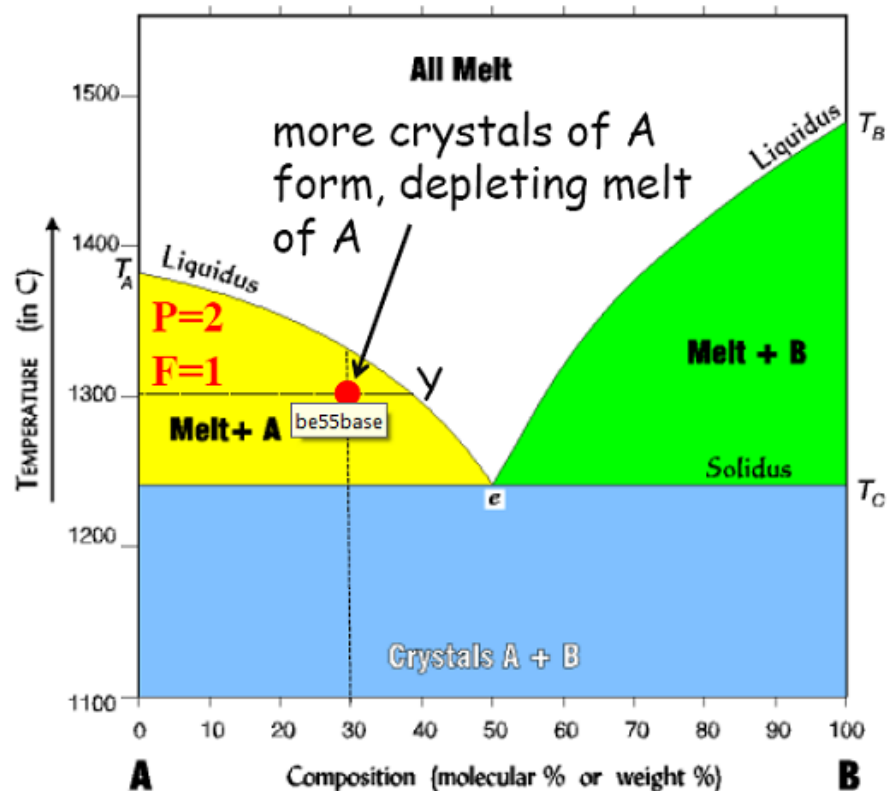
452

e.g., start
with liquid
at a certain
composition



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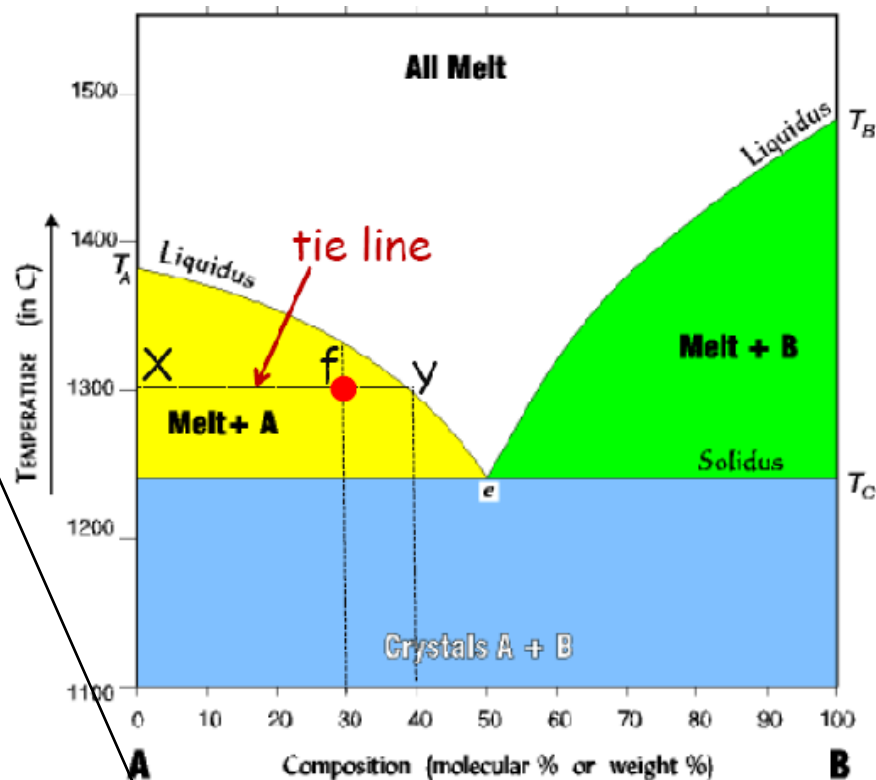
keep cooling



the system is now a mixture of crystals of pure A and the melt of composition y

455

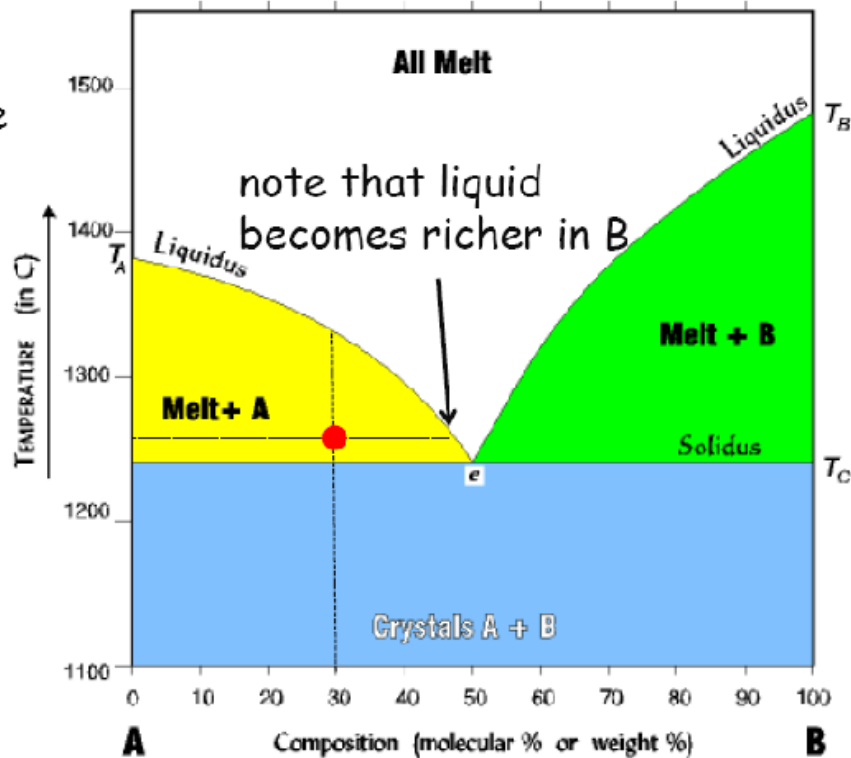
You will learn LEVER rule in the second document-chapter 9
Page -: 05, 9.7c



The relative amounts of the two phases ("phase composition") can be determined by using the **lever rule** along the **tie line** XfY:
Amount of A = fY/XY and amount of liquid = Xf/XY .

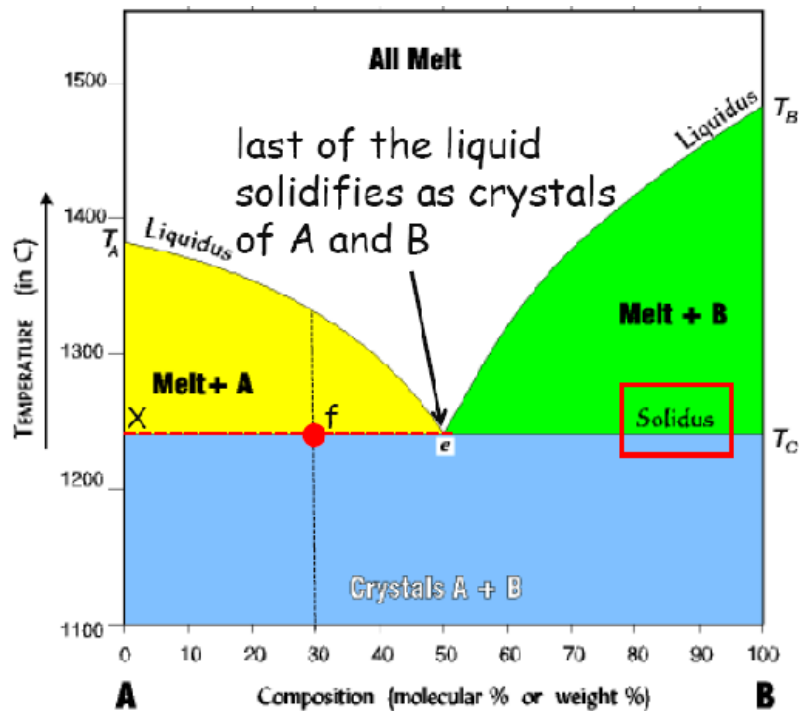
457

cool more



We can use the phase diagram to determine the **phase composition**, the *relative amounts* of A and melt at a certain T and bulk comp.

cool some more



passing through the solidus is called the **eutectic reaction**:



Solidus curve: gives the lowest temperature at which liquids can exist in equilibrium over a given compositional range

EUTECTIC COMPOSITION (“easily melted”)

system with the eutectic composition will melt/freeze **congruently** (i.e., completely, with no change in composition) and at the lowest T

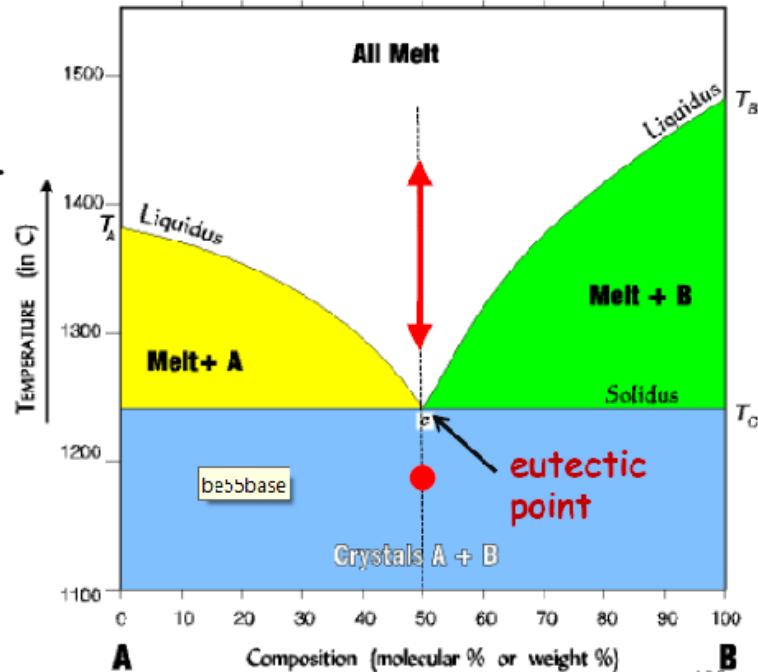
composition with the lowest melting point

eutectic point is invariant

$$P + F = 3$$

$$F = 0$$

- 3 phases present (A, B, liquid e)



THERMODYNAMICS OF LIQUID IMMISCIBILITY

Enthalpy of mixing determines whether two liquids A and B mix

$$\Delta G_{\text{mix}} = \Delta H - T\Delta S$$

- $\Delta S > 0$ for mixing
- if $\Delta H < 0$ (stronger A-B attraction), the liquids are miscible at all temps
- if $\Delta H > 0$ (stronger A-A, B-B attraction):
 - possibly miscible if $\Delta H < T\Delta S$ (high T)
 - immiscible when $\Delta H > T\Delta S$ (low T)

SOLID SOLUTIONS

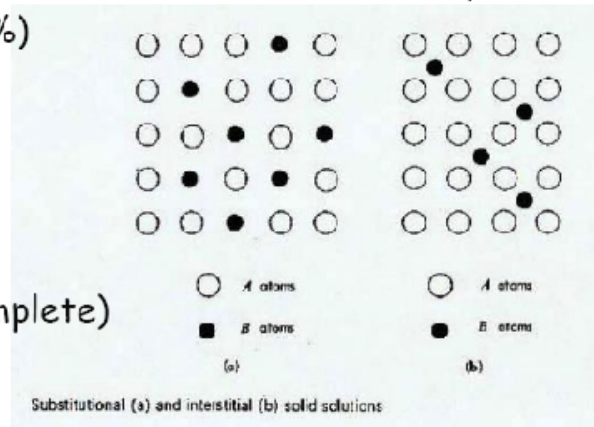
a solid-state solution; a crystalline material with variable composition

Two types:

- substitutional (solute atom/ion replaces solvent atom/ion)
- interstitial (solute atom/ion occupies interstitial site)

Formation of substitutional solid solutions favored by:

- similar atomic/ionic radii (15%)
- same crystal structure
- similar electronegativities
- same valency

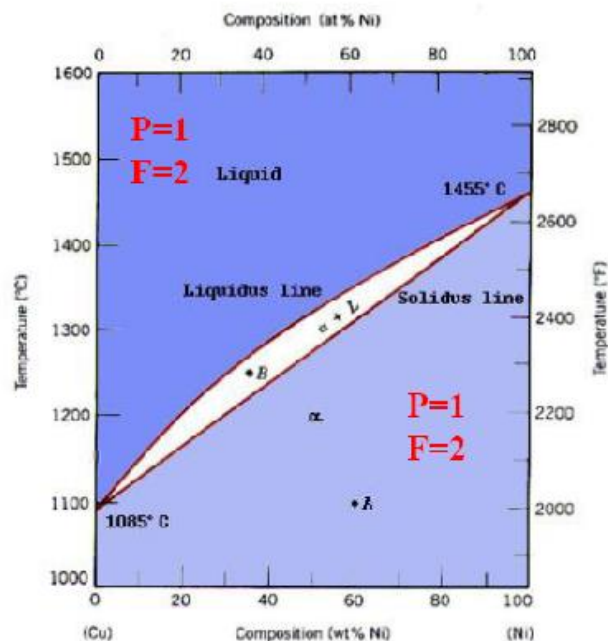


Examples:

- Al_2O_3 - Cr_2O_3 (substitutional, complete)
- Si-Ge (substitutional, complete)
- Fe-C (interstitial, partial)

BINARY SYSTEMS WITH SOLID SOLUTIONS

simplest solid solution system has complete miscibility in both solid and liquid states



- copper melts at 1085 C
- nickel melts at 1455 C
- above liquidus: single phase liquid
- between liq. and sol.: two phases
- below solidus: single phase solid a α -phase is a substitutional solid consisting of both Cu and Ni and on an FCC lattice

